

Noah Ruderman

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EDUCATION

San Jose State University

Aug. 2024 – Dec. 2027

Bachelor of Science in Computer Science

GPA: 3.96/4.00

Coursework: Data Structures and Algorithms, Operating Systems, Computer Architecture

Organizations: Competitive Programming Team, ACM at SJSU

Awards: 18th place, ICPC Pacific Northwest Regional; LeetCode Guardian

PROFESSIONAL EXPERIENCE

Student Assistant

Mar. 2026 – Present

San Jose State University

San Jose, CA

- Developed a web application for Dr. Ashwini Wagle that delivers culturally relevant nutrition data
- Implemented REST endpoints with FastAPI to support frontend-backend communication
- Designed and maintained a PostgreSQL database for nutrition data and user meal logs
- Set up pytest CI testing to identify backend API regressions, catching multiple issues before deployment
- Containerized the application with Docker to standardize development and testing environments

FIRST Robotics

Aug. 2020 – May 2024

Monta Vista Robotics Team

Cupertino, CA

- Prototyped, machined, and assembled mechanical components for the team's competition robot
- Designed and modeled a training car to introduce new members to the build process
- Led CAD lessons for students in grades 2–9, teaching core modeling concepts

Building Attendant

Oct. 2023 – Present

City of Cupertino

Cupertino, CA

- Provided AV support during events by troubleshooting technical issues
- Coordinated facility rentals and provided customer service to guests

PROJECTS

Online Board Game | *TypeScript, WebGL, Go, Render*

Feb. 2024 – Present

- Built a full-stack web application with a Go backend, WebSocket API, and WebGL2 frontend
- Implemented real-time multiplayer support for multiple devices connected to a shared game session
- Deployed on Render for global public access

Terrain Viewer | *C++, CMake, TypeScript, WebGPU*

Dec. 2024 – Jan. 2025

- Developed an application for analyzing Earth elevation data
- Built a WebGPU-powered web interface
- Implemented multiple visualization modes to explore topographic features around a selected location

Custom CPU Design | *C++, Java, Assembly*

Jan. 2023 – Aug. 2023

- Designed and emulated an 8-bit CPU instruction set architecture
- Implemented an interrupt controller and MMIO interface for graphical output
- Built an assembler to streamline development for the custom CPU

Discord Bot | *JavaScript, Node.js, Discord.js, API*

Feb. 2020 – Nov. 2023

- Built and maintained a Discord bot for 20+ active users
- Developed persistent multiplayer mini-games and text/image commands

TECHNICAL SKILLS

Languages: Java, C, C++, Rust, Python, Go, JavaScript, TypeScript, Bash, SQL

Frameworks: Node.js, JUnit, FastAPI, React

Developer Tools: Git, Linux/Unix, CMake, Docker, CI/CD

Databases: PostgreSQL, MySQL

Libraries: NumPy, Matplotlib, PyTorch, pytest